Arista Access Node MAR-ACC-04 QoS Congestion Event Resolution SOP

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1923

Description

This SOP is developed to address a minor level QoS congestion event on the Arista Access Node MAR-ACC-04 at Marathahalli Core. The failure domain in question revolves around microbursts, interface tail drops, and QoS queue starvation, which are inherent consequences of the Arista EOS's (Extensible Operating System) default queueing mechanism on Access nodes. Specifically, the issue is caused by the default configuration of the egress queue, which does not account for the varying packet sizes and burst rates that traverse the network. When the egress queue becomes congested, it leads to microbursts of packets, which can cause interface tail drops and subsequently result in QoS queue starvation. This congestion event is further exacerbated by the lack of QoS policies in place to regulate traffic flow on the Access node.

Pre-requisite Checks

- check: Verify that the Marathahalli Core site is accessible and that all necessary equipment is within reach., action: Dispatch remote hands to the site if necessary.

- check: Check the redundant link status to ensure that the secondary link is functioning correctly., action: Notify NOC if the secondary link is down.

- check: Verify that the Access node's power supply is stable and not causing any issues., action: Notify NOC if the power supply is unstable.

Potential Root Causes

Cause	Probability
Default egress queue configuration not accounting for varying packet sizes and burst rates.	High
Lack of QoS policies in place to regulate traffic flow.	Medium
Microbursts of packets causing interface tail drops.	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface egress-queue detail, Expected Output: Egress queue configuration and statistics., Description: This command displays the egress queue configuration and statistics, including the queue depth, packet drop count, and other relevant information. It helps to identify if the egress queue is congested and if it is causing packet drops.

Command: show qos policy, Expected Output: QoS policy configuration and statistics., Description: This command displays the QoS policy configuration and statistics, including the policy name, class map,
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and queueing configuration. It helps to identify if there are any QoS policies in place to regulate traffic flow on the Access node.

Command: show queue statistics, Expected Output: Queue statistics, including packet drop count and queue depth., Description: This command displays the queue statistics, including the packet drop count and queue depth. It helps to identify if the queues are congested and if it is causing packet drops.
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Resolution Steps

Step 1: Step: Configure egress queue with increased buffer size., Command: config terminal queue egress-queue buffer-size 10000 end, Description: This step configures the egress queue with an increased buffer size to account for varying packet sizes and burst rates. It helps to prevent microbursts of packets and interface tail drops.

Step 2: Step: Apply QoS policy to regulate traffic flow., Command: config terminal policy qos policy-name queueing configuration end, Description: This step applies a QoS policy to regulate traffic flow on the Access node. It helps to prevent congestion and packet drops by prioritizing traffic based on classification and marking.

Step 3: Step: Verify egress queue configuration and statistics., Command: show interface egress-queue detail, Description: This step verifies the egress queue configuration and statistics to ensure that the congestion event has been resolved.

Post-Check Verification

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Command: show interface egress-queue detail, Expected Output: Egress queue configuration and statistics showing no congestion., Description: This command verifies that the egress queue configuration and statistics show no congestion and no packet drops.

Command: show qos policy, Expected Output: QoS policy configuration and statistics showing correct policy application., Description: This command verifies that the QoS policy configuration and statistics show correct policy application and no congestion.

Command: show queue statistics, Expected Output: Queue statistics showing no packet drops and correct queue depth., Description: This command verifies that the queue statistics show no packet drops and correct queue depth.
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Roll-back Procedures

- Step: Revert egress queue configuration to previous state., Command: config terminal undo queue egress-queue buffer-size 10000 end, Description: This step reverts the egress queue configuration to its previous state in case the fix fails.

- Step: Revert QoS policy configuration to previous state., Command: config terminal undo policy qos policy-name queueing configuration end, Description: This step reverts the QoS policy configuration to its previous state in case the fix fails.